

navya@kali: ~

Session Actions Edit View Help

(navya@kali)-[~]

\$ ifconfig

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fe8a:3253 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8a:32:53 txqueuelen 1000 (Ethernet)
    RX packets 6 bytes 2028 (1.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 26 bytes 3818 (3.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
```

```
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 196 bytes 11720 (11.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 196 bytes 11720 (11.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

(navya@kali)-[~]

\$ nmap -sn 192.168.56.104/24

Starting Nmap 7.98 (<https://nmap.org>) at 2026-06-11 11:50 +0530

Nmap scan report for 192.168.56.1

Host is up (0.0012s latency).

MAC Address: 0A:00:27:00:00:0A (Unknown)

Nmap scan report for 192.168.56.100

Host is up (0.00058s latency).

MAC Address: 08:00:27:BE:B5:1F (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.56.104

Host is up.

Nmap done: 256 IP addresses (3 hosts up) scanned in 1.95 seconds

(navya@kali)-[~]

\$

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Session Actions Edit View Help

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```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
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(navya@kali)-[~]

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Starting Nmap 7.98 (<https://nmap.org>) at 2026-06-11 11:50 +0530

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MAC Address: 0A:00:27:00:00:0A (Unknown)

Nmap scan report for 192.168.56.100

Host is up (0.00058s latency).

MAC Address: 08:00:27:BE:B5:1F (Oracle VirtualBox virtual NIC)

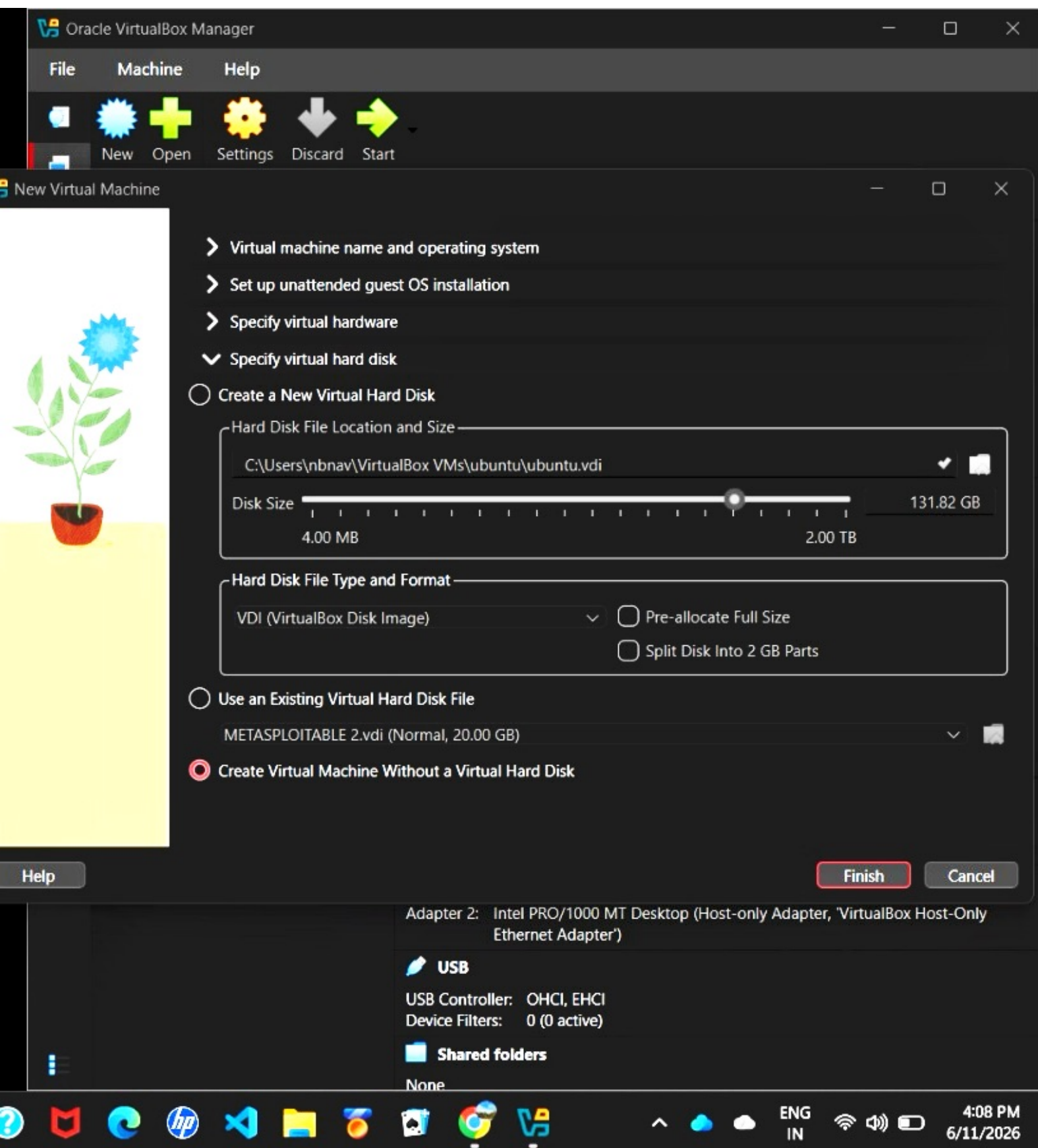
Nmap scan report for 192.168.56.104

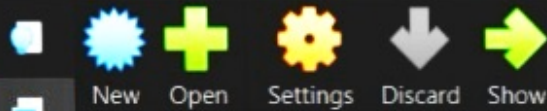
Host is up.

Nmap done: 256 IP addresses (3 hosts up) scanned in 1.95 seconds

(navya@kali)-[~]

\$





Kali Linux
Powered Off

METASPLOITABLE 2
Powered Off

ubuntu
Running

Details

General

Name: ubuntu
Operating System: Ubuntu 25.04
(Plucky Puffin) (64-bit)

System

Base Memory: 3878 MB
Processors: 3

Preview



ubuntu [Running] - Oracle VirtualBox

```
[ 7.257992] async_tx: api initialized (async)
done.
Begin: Running /scripts/init-premount ... done.
Begin: Mounting root file system ... Begin: Running /scripts/nfs-top ... done.
Begin: Running /scripts/nfs-premount ... done.
[ 7.861215] loop0: detected capacity change from 0 to 6684880
[ 8.018677] loop1: detected capacity change from 0 to 1157752
[ 8.202081] loop2: detected capacity change from 0 to 1577296
[ 8.278961] overlaysfs: null uuid in non-single lower fs '/', falling back to
xino=off,index=off,nfs_export=off.
Begin: Running /scripts/casper-premount ... done.
done.
Begin: Creating debconf-communicate fifo mechanism ... [ 9.899752] evm: overl
ay not supported
done.
Begin: Running /scripts/casper-bottom ... Begin: Moving mount points... ... done
.
Begin: Configuring fstab... ... done.
Begin: Setting up swap... ... done.
Begin: Setting up locales... ... done.
Begin: Setting up automatic login... ... done.
Begin: Setting hostname... ... done.
Begin: Setting up console keyboard... ... done.
Begin: Applying desktop settings... ...
```

9-9c4b-

o (6.07 GB)

Right Ctrl

USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)

Shared folders

None



ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
ed")
dbus-daemon[553]: [session uid=1000 pid=18446744073709551615 pidfd=5] Successful
ly activated service 'org.gtk.vfs.Daemon'
dbus-daemon[553]: [session uid=1000 pid=18446744073709551615 pidfd=5] Activating
service name='org.gtk.vfs.Metadata' requested by ':1.0' (uid=1000 label="unconf
ined")
fuse: device /dev/fuse not found. Kernel module not loaded?
dbus-daemon[553]: [session uid=1000 pid=18446744073709551615 pidfd=5] Successful
ly activated service 'org.gtk.vfs.Metadata'
A connection to the bus can't be made
done.
Begin: Setting up init... .. done.
Begin: Configuring accessibility options... .. done.
Begin: Disabling update-notifier... .. done.
Begin: Enabling detection of crashes... .. done.
Begin: Disabling unnecessary KDE services... .. done.
Begin: Disabling the lock screen for LXQt (LP: #2000787)... .. done.
Begin: Generating APT cache for cdrom ... Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:4 file:/cdrom resolute/restricted amd64 Packages [14.5 kB]
Get:5 file:/cdrom resolute/main amd64 Packages [31.7 kB]
```

Right Ctrl

Jun 12 12:36

vboxuser@Ubuntu: ~

```
vboxuser@Ubuntu:~$ sudo nmap -p22,80 --script banner 192.168.56.103
[sudo: authenticate] Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-12 12:36 +0000
Nmap scan report for 192.168.56.103
Host is up (0.0011s latency).
```

PORT	STATE	SERVICE
22/tcp	filtered	ssh
80/tcp	filtered	http

```
Nmap done: 1 IP address (1 host up) scanned in 2.70 seconds
vboxuser@Ubuntu:~$
```

Terminal

ENG

6:06 PM

Jun 12 12:51

```
vboxuser@Ubuntu: ~
```

```
vboxuser@Ubuntu:~$ sudo nmap -F -T4 --script banner 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-12 12:50 +0000
Nmap scan report for 192.168.56.103
Host is up (0.020s latency).
Not shown: 82 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
|_banner: 220 (vsFTPd 2.3.4)
22/tcp    open  ssh
|_banner: SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
23/tcp    open  telnet
|_banner: \xFF\xFD\x1B\xFF\xFD \xFF\xFD#\xFF\xFD'
25/tcp    open  smtp
|_banner: 220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
53/tcp    open  domain
80/tcp    open  http
Trash     open  rpcbind
111/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
513/tcp   open  login
514/tcp   open  shell
|_banner: \x01getnameinfo: Temporary failure in name resolution
2049/tcp   open  nfs
2121/tcp   open  ccproxy-ftp
3306/tcp   open  mysql
|_banner: >\x00\x00\x00\x0A5.0.51a-3ubuntu5\x00\x09\x00\x00\x00|TnNy9MV\x
|_00,\xAA\x08\x02\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00...
5432/tcp   open  postgresql
5900/tcp   open  vnc
|_banner: RFB 003.003
6000/tcp   open  X11
8009/tcp   open  ajp13

Nmap done: 1 IP address (1 host up) scanned in 18.17 seconds
vboxuser@Ubuntu:~$
```


Reconnaissance - preparing info for attacks.

* Active - we interact with the target directly

* passive - we get the info that is publically available using tools.

Tools -

1. whois

eg. whois lynx-infosec.com

2. dig - gives ip address of target

eg. dig lynx-infosec.com

3. whatweb

eg. whatweb --help

whatweb bzu.edu.pk

whatweb lynx-infosec.com

4. nslookup - tells us the name servers

eg. nslookup

> cisco.com

set type = NS

set type = MX (Mail exchange)

set type = SOA

5. wafw00f (tells firewall)

eg. wafw00f nust.edu.pk

6. the Harvester -h (help) (email, sub domains)

-d (domain)

-b (source)

-l (limit)

-s (shiden)

Google Dorking

★ Finding exposed login pages

inurl: "/admin/login" site:.in

★ Finding database files

filetype:sql "INSERT INTO" "users"

★ Finding password files

filetype:txt intext:"username" intext:"password"

★ Finding webcams and devices

intitle:"webcam XP 5" inurl:8080

How to protect -

1. use robots.txt properly
2. Do not leave sensitive files publically accessible.
3. Disable directory listing
4. Perform regular google dork audits
5. Use google search console

Ping Sweep -

used to discover active hosts on an IP network

- Identify active devices on a network
nmap -sn 192.168.XX.XX/24

Banner Grabbing (using ubuntu)

```
sudo nmap -p 22, 80 --script banner 192.168.56.1
```

```
sudo nmap -F -T4 --script banner 192.168.56.1
```

used to collect info about a service running on open port such as -

- Service name
- Software version
- OS details